Lowest Common Sequence

```c
#include<stdio.h>
#include<string.h>

int max(int a, int b) {
    return (a > b) ? a : b;
}

int lcs(char *a,char *b)
{
    int m = strlen(a);
    int n = strlen(b);
    if(m<n)
    {
        int lcs[m+1][n+1];
        for(int i=0;i<=m;i++)
        {
            lcs[i][0]=0;
        }
        for(int j=0;j<=n;j++)
        {
            lcs[0][j]=0;
        }
        for(int i=1;i<=m;i++)
        {
            for(int j=1;j<=n;j++)
            {
                if(a[i-1]==b[j-1])
                {
                    lcs[i][j]=1+lcs[i-1][j-1];
                }
                else
                {
                    lcs[i][j]=max(lcs[i-1][j],lcs[i][j-1]);
                }
            }
        }
        for (int i = 0; i <= m; i++) {
            for (int j = 0; j <= n; j++) {
                printf("%d ", lcs[i][j]);
            }
            printf("\n");
        }
        return lcs[m][n];
    }
    else
    {
```

```
        int lcs[n+1][m+1];
        for(int i=0;i<=n;i++)
        {
            lcs[i][0]=0;
        }
        for(int j=0;j<=m;j++)
        {
            lcs[0][j]=0;
        }
        for(int i=1;i<=n;i++)
        {
            for(int j=1;j<=m;j++)
            {
                if(b[i-1]==a[j-1])
                {
                    lcs[i][j]=1+lcs[i-1][j-1];
                }
                else
                {
                    lcs[i][j]=max(lcs[i-1][j],lcs[i][j-1]);
                }
            }
        }
        for (int i = 0; i <= n; i++) {
            for (int j = 0; j <= m; j++) {
                printf("%d ", lcs[i][j]);
            }
            printf("\n");
        }
        return lcs[n][m];
    }
}

int main()
{
    int num1,num2;
    char a[100];
    char b[100];
    printf("Enter Your 1st String:");
    scanf("%s", a);
    printf("\nYour 1st String Is:\n");
    printf("%s", a);
    printf("\nEnter Your 2nd String:");
    scanf("%s", b);
    printf("\nYour 2nd String Is:\n");
    printf("%s\n", b);
    int length=lcs(a,b);
    printf("\nLength of Longest Common Subsequence: %d\n", length);
```

```
}
```

Output:

```
Enter Your 1st String:kush

Your 1st String Is:
kush
Enter Your 2nd String:kushagra

Your 2nd String Is:
kushagra
0 0 0 0 0 0 0 0 0
0 1 1 1 1 1 1 1 1
0 1 2 2 2 2 2 2 2
0 1 2 3 3 3 3 3 3
0 1 2 3 4 4 4 4 4

Length of Longest Common Subsequence: 4
PS C:\Users\KUSHAGRA\.vscode\sem 3>
```